

CUSUM RAP – User Help (WS5.WP1.2)

This document explains, step by step:

1. How to **configure the parameters** of the CUSUM RAP before running it.
2. How to **run** the CUSUM RAP in R / RStudio.

It assumes you already have the repository structure:

```
WP1.2_CUSUM_RAP/  
  R/  
    01_functions_cusum.R  
    02_functions_plot.R  
    03_functions_io.R  
  input/  
  output/  
  main.R  
  README.md  
  HELP_EN.md  
  HELP_ES.md
```

and that your aggregated input file from API-Pop is a CSV inside `input/`.

1. Configuration – How to set the parameters before running

All user-configurable parameters are located **at the top of `main.R`**.

Open `main.R` in RStudio and look for this block:

```
## -----  
## 1) User-configurable parameters  
## -----  
  
# Geographic analysis level  
# Options: "country", "province", "department", "censal_fraction"  
location_level <- "province"  
  
# CUSUM threshold h (to be calibrated)  
cusum_h <- 3
```

```

# Baseline length (in weeks):
#   - detection period = last baseline_length_weeks
#   - baseline period  = all weeks before that
baseline_length_weeks <- 52

# Input path: API-Pop aggregated CSV
input_path <- "input/rumor_counts_by_week_unit.csv"

```

You can edit the values on the right-hand side ("province", 3, 52, etc.) according to your needs.

1.1 Parameter: location_level

```
location_level <- "province"
```

This parameter controls **the spatial level of analysis**. The RAP will aggregate weekly counts to this level before running CUSUM.

Allowed values:

- "country"
All data aggregated at **country level**.
- "province"
Data aggregated by country + province.
- "department"
Data aggregated by country + province + department.
- "censal_fraction"
Data aggregated by country + province + department + censal_fraction.

Examples:

- `location_level <- "province"` → CUSUM is run **per province**.
- `location_level <- "censal_fraction"` → CUSUM is run **per censal fraction** (most granular level).

Make sure your input CSV has the corresponding location columns (country, province, department, censal_fraction). These column names can be adapted inside `prepare_weekly_data_geo()` if needed.

For the time dimension, the RAP accepts **two alternative formats**:

- Either **separate year and week columns** (e.g. year = 2024, week = 5), or

- A **single combined week column** with values in "year-week" format (e.g. "2024-05" or "2024-5").

Internally, the helper function `prepare_weekly_data_geo()` parses this information and creates the `year`, `week`, `epi_date`, and `time_index` variables used by the CUSUM analysis. The mapping between input column names and internal variables can be configured via the arguments `col_year`, `col_week`, and `col_yearweek` in `prepare_weekly_data_geo()`.

1.2 Parameter: `cusum_h`

```
cusum_h <- 3
```

- `cusum_h` is the **CUSUM decision threshold**.
- When the CUSUM statistic `S_t` for an analysis unit reaches or exceeds `h`, the RAP flags an **alarm** for that week.

This value should be **calibrated** according to:

- Desired false alarm rate.
- Sensitivity to changes in rumor frequency.

For testing and development, you can start with:

- `cusum_h <- 3` or `cusum_h <- 4`

and later adjust based on simulation studies or expert calibration.

1.3 Parameter: `baseline_length_weeks`

```
baseline_length_weeks <- 52
```

This parameter controls how the RAP **splits the time series** into:

- **Baseline period:** used to estimate expected counts (baseline model).
- **Detection period:** where CUSUM is applied.

For each analysis unit:

- Let `T` be the maximum `time_index` (last available week).
- Detection period = **last `baseline_length_weeks` weeks**:
weeks with `time_index` > `T - baseline_length_weeks`
- Baseline period = **all weeks before that**:
weeks with `time_index` <= `T - baseline_length_weeks`

Example:

- If `baseline_length_weeks` <- 52 and you have 200 weeks of data:
 - Weeks 1–148 → baseline (used to fit the Poisson GLM).
 - Weeks 149–200 → detection (CUSUM applied).

If you change to `baseline_length_weeks` <- 26:

- Weeks 1–174 → baseline.
- Weeks 175–200 → detection.

You can increase the baseline length if you have long historical series.

1.4 Parameter: `input_path`

```
input_path <- "input/rumor_counts_by_week_unit.csv"
```

This defines **where the RAP reads the input CSV**.

- By default, it assumes a file named `rumor_counts_by_week_unit.csv` inside the `input/` folder.
- You can change the filename if needed, for example:

```
input_path <- "input/api_pop_output_2025.csv"
```

The file must include the spatial location columns (`country`, `province`, `department`, `censal_fraction`), the weekly counts column (e.g. `n_cases`), and the time information in one of the two supported formats:

- Separate `year` and `week` columns, or
- A single combined `week` column with "year-week" format.

2. How to Run the RAP – Step by step in RStudio

2.1 Open the project in RStudio

1. Start RStudio.
2. Click **File** → **Open Project...**
3. Navigate to the WP1.2_CUSUM_RAP/ folder and open the `.Rproj` file (if it exists) or select the folder as a project.

Check that you can see the subfolders `R/`, `input/`, `output/` in the **Files** pane.

2.2 Install required R packages (first time only)

In the **Console**, run:

```
install.packages(c(
  "dplyr", "ggplot2", "readr",
  "lubridate", "ISOweek", "tidyr", "rlang"
))
```

You only need to do this **once per machine**.

2.3 Place the input CSV in input/

1. Copy your aggregated API-Pop CSV file into the `input/` folder.
2. Make sure the filename and columns match what you configured in:
 - `input_path` in `main.R`, and
 - `prepare_weekly_data_geo()` in `R/03_functions_io.R`.

Example:

`input/rumor_counts_by_week_unit.csv`

2.4 Configure parameters in main.R

1. In the **Files** pane, click on `main.R` to open it.
2. At the top of the file, review and adjust:

```
location_level <- "province"
cusum_h <- 3
baseline_length_weeks <- 52
input_path <- "input/rumor_counts_by_week_unit.csv"
```

3. Save changes (**Ctrl+S** or **Cmd+S**).

2.5 Run the RAP

With `main.R` open, run the full script:

- Option A: Press **Ctrl + Shift + Enter** (Cmd + Shift + Enter on Mac).
- Option B: Click the **Source** button at the top-right of the script editor.

R will:

1. Load the functions (`source("R/...")`).
2. Read the input CSV from `input_path`.
3. Prepare weekly data and geographic aggregation at the selected `location_level`.
4. Fit baseline models and compute CUSUM for each analysis unit.
5. Save outputs in the `output/` folder.

If an error occurs, read the message in the **Console**. The most common issues are:

- Missing R packages → install them.
- Incorrect `input_path` → fix the file name or location.
- Missing or misnamed columns in the CSV → adapt column names or `prepare_weekly_data_geo()`.

2.6 Check the outputs

After running `main.R`, look inside the `output/` folder. You should find:

- A CSV file with CUSUM results, for example:
`cusum_results_province.csv`
(the suffix matches the chosen `location_level`).
- Example plots:
 - `example_province_series.png`
→ observed vs expected counts + alarms for one example unit.
 - `example_province_cusum.png`
→ CUSUM process for the same unit.
 - `cusum_alarms_overview_province.png`
→ tile/heatmap of alarms across all units and weeks.

You can open the PNG files with any image viewer or directly from RStudio.

3. Contact / Support

If you encounter persistent errors, please share:

- The full error message from the R console.
- A short description of the input data (number of rows, columns).
- The values you used for `location_level`, `cusum_h`, `baseline_length_weeks`.

This will help the development team diagnose and fix the issue.